DIP Aim and Theory

Experiment 1 – Create a program to demonstrate Geometric transformations - image rotation, scaling, translation.
https://colab.research.google.com/drive/1hwKFaFWLsCR9k9lEskao4C_WvdFnQgtb?usp=sharing

Geometric transformations modify the spatial arrangement of image pixels without changing their intensity values. Common transformations include translation, scaling, and rotation. Translation shifts an image to a new position, scaling changes its size, and rotation turns the image around a fixed point. These operations are widely used in image alignment, object tracking, image registration, and computer vision applications for improving image interpretation and analysis.

Experiment 2 – Display of FFT (1-D & 2-D) of an image and apply two-dimensional fourier transform using DFT and masking with DFT
https://colab.research.google.com/drive/1hKeb-ToS-AIgkW-l7_O50RqWT54WRRSx?usp=sharing

The Fourier Transform converts an image from the spatial domain into the frequency domain, where image details are represented as frequency components. The Discrete Fourier Transform (DFT) and Fast Fourier Transform (FFT) are used to analyze image frequencies efficiently. Applying masks in the frequency domain helps perform operations such as smoothing, sharpening, and noise reduction. These techniques are important in image enhancement, filtering, and pattern analysis applications.

Experiment 3 – Write a program for Contrast stretching of a low contrast image, Histogram, Histogram Equalization, and display of bit planes
https://colab.research.google.com/drive/1GZggzgpAJtUv39ApooAejCxHiqB7ygKB?usp=sharing

Contrast stretching and histogram processing are image enhancement techniques used to improve image quality and visibility. Contrast stretching expands the intensity range of low-contrast images, making details clearer. A histogram represents the distribution of pixel intensities, while histogram equalization redistributes intensity values for better contrast. Bit-plane slicing separates image pixels into binary layers, helping analyze significant visual information and noise components in digital images.

Experiment 4 – Computation of Mean, Standard Deviation, Correlation coefficient of the given Image.
https://colab.research.google.com/drive/1lE7jwRrCbXsY9sbnyO3WoRN9lgs_gpuZ?usp=sharing

Statistical measures such as mean, standard deviation, and correlation coefficient help analyze image characteristics. The mean represents average brightness, while standard deviation indicates contrast and intensity variation. The correlation coefficient measures similarity between images or pixel patterns. These parameters are useful in image analysis, feature extraction, and pattern recognition. They provide quantitative information about image quality, texture, and relationships between different image datasets.

Experiment 5 – Implementation of Image Smoothening Filters (Mean and Median filtering of an image)
https://colab.research.google.com/drive/19VMHQ4_Lo16pVCb93A5s7cw-snm3-BvW?usp=sharing

Image smoothening filters are preprocessing techniques used to reduce noise and unwanted intensity variations in images. The Mean filter replaces each pixel with the average value of neighboring pixels, effectively reducing Gaussian noise but causing some blurring. The Median filter replaces a pixel with the median of surrounding values, preserving edges while removing salt-and-pepper noise. Smoothening improves image quality and supports better segmentation and feature extraction.

Experiment 6 – Implementation of Image Sharpening Filters and Edge Detection using Gradient Filters.
https://colab.research.google.com/drive/1BhueEQakk6eGNkKpXZZ1Y-QIHZp-UrMj?usp=sharing

Image sharpening enhances edges and fine details by emphasizing high-frequency components in an image. Gradient-based edge detection methods identify boundaries by detecting rapid intensity changes. Operators such as Sobel, Prewitt, and Roberts calculate image gradients in horizontal and vertical directions. Sharpening improves visual clarity, while edge detection extracts structural information for further analysis. These techniques are widely used in computer vision, segmentation, and object recognition systems.

Experiment 7 – Implementation of Image Compression by DCT, DPCM, HUFFMAN coding
https://colab.research.google.com/drive/1BhueEQakk6eGNkKpXZZ1Y-QIHZp-UrMj?usp=sharing

Image compression reduces image storage size while maintaining acceptable visual quality. The Discrete Cosine Transform (DCT) converts image data into frequency components, allowing less important details to be removed. Differential Pulse Code Modulation (DPCM) compresses images by encoding differences between neighboring pixels. Huffman coding further reduces data size using variable-length codes based on symbol frequency. These techniques are commonly used in multimedia systems and JPEG image compression standards.

Experiment 8 – To implement and study various image restoration techniques including Inverse Filtering, Wiener Filtering, and Constrained Least Squares Filtering in Python.
https://colab.research.google.com/drive/1ucbNw1_ySt-j09m1cHUQGjKq97XEnDa1?usp=sharing

Image restoration techniques aim to recover degraded images affected by blur or noise. Inverse filtering restores images by reversing the degradation function but is sensitive to noise. Wiener filtering improves restoration by considering noise statistics and minimizing mean square error. Constrained Least Squares (CLS) filtering introduces smoothness constraints for balanced restoration. These methods enhance image quality and are widely used in medical imaging, satellite imaging, and photography applications.

Experiment 9 – To implement intensity slicing (gray-level slicing) techniques in Python for image enhancement and study the effect of highlighting specific intensity ranges.
https://colab.research.google.com/drive/1TsqdSnsWEhpPBCLOlG8IzBcGZMl7x0-t?usp=sharing

Intensity slicing is an image enhancement method used to highlight specific ranges of pixel intensities. It emphasizes important gray-level regions while suppressing less relevant areas. The technique can preserve the background or isolate only selected intensity ranges. Gray-level slicing improves visibility of significant features in images and is commonly used in medical imaging, satellite image analysis, and object detection where certain intensity values represent meaningful information.

Experiment 10 – To study and implement the Canny Edge Detection Algorithm in Python and compare it with Sobel, Prewitt, Roberts, and Laplacian of Gaussian (LoG) edge detection methods.
https://colab.research.google.com/drive/1c3eOF2K8nzwTshu0rHk9VBLTz3K9TVnO?usp=sharing

The Canny Edge Detection algorithm is a multi-stage method used to detect edges accurately while minimizing noise effects. It includes Gaussian smoothing, gradient calculation, non-maximum suppression, and hysteresis thresholding. Other edge detection methods such as Sobel, Prewitt, Roberts, and Laplacian of Gaussian (LoG) also identify intensity changes using different operators. Comparing these methods helps evaluate edge quality, computational efficiency, and noise sensitivity in image processing tasks.